



**United
Nations**

Anticipatory Action Framework

Fiji Tropical Cyclones – 2023

Approved by the RC on 3 November 2023

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1. Introduction

Pacific Island countries are among the most vulnerable to natural hazards exacerbated by climate change, and there is a growing interest in scaling up anticipatory action in the region to support vulnerable communities before a hazard strikes.

Following several years of piloting, OCHA is mainstreaming and scaling-up anticipatory action. In its 2023-2026 Strategic Plan, OCHA commits to “support and facilitate a systemic shift to coherent and embedded anticipatory approaches, including anticipatory action. OCHA will use its own financing tools to facilitate, generate evidence for and scale up collective anticipatory action.” OCHA implements this strategy – inter alia – by facilitating coordinated, collective anticipatory action frameworks.

In March 2023 the Pacific Anticipatory Action Week brought together 15 governments and national/ regional development partners in Nadi, Fiji. Following these discussions, Fiji was identified as a potential pilot for coordinated anticipatory action financed through OCHA-managed pooled funds.

After close consultations with the Government and operational partners, the United Nations Resident Coordinator for Fiji formally requested UN support for collective anticipatory action for tropical cyclones in Fiji. Two weeks later, the Under-Secretary-General for Humanitarian Affairs and Emergency Relief Coordinator approved the request, including pre-arranged CERF financing, encouraging the United Nations in country to collaborate closely with the government; to build on existing systems; to identify co-financing; and to document the impact of the pilot. Furthermore, the partners agreed to build on the existing resilience of vulnerable populations.

This pilot builds on existing disaster risk management capacity of Fiji, including well established climate and disaster risk governance, early warning systems and coordination structures. The need for enhanced early warning and response to a range of hazards, including tropical cyclones, has been identified as priority in national policy settings, including the National Adaptation Plan and National DRR policy. Fiji's recent national midterm review report on the implementation of the Sendai¹ Framework for DRR also identified the need for strengthened early warning system in the country, which further supports its inclusion as a pilot country under the Early Warning for All initiative.

In line with best practices for anticipatory action to minimize and address the impact of sudden onset climate-related disasters, this framework for coordinated anticipatory action to tropical cyclones in Fiji pre-agrees when and on what basis there will be a trigger ahead of a specific storm; how much pre-arranged financing will go to whom; and for what pre-agreed activities the funding will be used for. This framework also includes provisions to document the impact of the pilot.

Coordination with local authorities

The National Disaster Management Act of 1998, supported by the National Disaster Management Office (NDMO), provides a comprehensive legal framework for disaster management, emphasizing prevention, preparedness, response, and recovery. The Early Warning System (EWS) plays a critical role in forecasting and monitoring natural hazards, ensuring timely alerts to at-risk communities. Moreover, Fiji's efforts in community-based disaster risk reduction and climate change adaptation programs underscore the government's dedication to proactive disaster risk management.

The anticipatory action framework has been developed in close partnership with the Fiji government, under the leadership of the NDMO in September 2023. The key components of the framework have been endorsed by the National Disaster Management Council. The trigger mechanism and activation protocol have been designed in collaboration with the Fiji Meteorological Service (FMS) and NDMO by building on existing forecasting and Early Warning Systems (EWS). Activities are streamlined through the national cluster system. Each partner agency collaborates closely with their respective national cluster co-leads, with a concerted effort to implement activities through established government systems wherever possible.

¹ ndmo.gov.fj/wp-content/uploads/2023/05/Fiji-Sendai-Mid-Term-Review-Report-2023.pdf

United Nations Office for Disaster Risk Reduction (UNDRR) supported the development of the Framework through facilitating engagement with key government stakeholders, including NDMO and FMS, and ensured the Framework is aligned with the existing disaster management system.



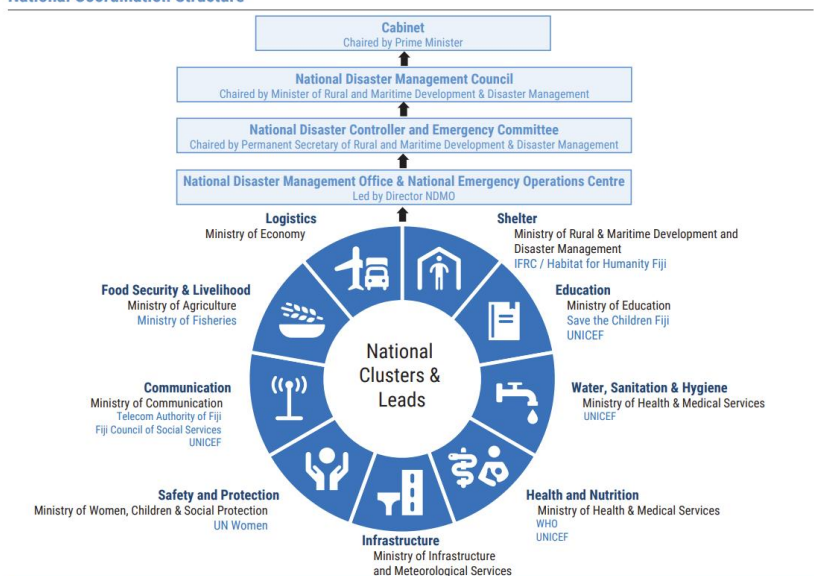
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Humanitarian Coordination Structure

As of 02 November 2023

The Humanitarian Coordination Structure in Fiji comprises of three levels: at the strategic level, the National Disaster Management Council chaired by the Minister for Rural and Maritime Development & Disaster Management (MRMDDM), at the coordination level, are three sub-committees; Mitigation & Prevention, Preparedness and Emergency committees chaired by the Permanent Secretary for MRMDDM - who is also the Disaster Controller. At the operational level, the National Emergency Operations Centre (NEOC) is managed by the National Disaster Management Office (NDMO). The legislative framework for the coordination of humanitarian affairs (preparedness, response, early recovery) is the National Disaster Management Act 1998. The national cluster system is based on the global cluster system which was introduced in 2005 during a major reform of the humanitarian coordination, known as the Humanitarian Reform Agenda, to enhance predictability, accountability and partnership. National clusters are led by relevant ministries or government departments and in few cases co-led by international organizations. The national clusters are being supported by the Pacific Humanitarian Team's regional clusters.

National Coordination Structure



Location Map



The boundaries and names shown and the designations used on this map do not imply official endorsement or acceptance by the United Nations.
Creation date: 02/11/2023 Sources: Government of Fiji OCHA/ReliefWeb Feedback: ochaapacific@un.org www.unocha.org www.reliefweb.int

■ National Cluster
■ National Lead Ministry
■ Cluster Co-lead

Risk/Hazard

Fiji is one of the most vulnerable countries in the world to tropical cyclones with significant impact on infrastructure, the economy, and the Fijian people, displacing thousands, creating humanitarian suffering, and causing loss of life.

Since 1969, there have been 33 TCs (Category 1 or higher) that have passed within 10 km of Fiji, and 16 that made landfall. Some of the most devastating cyclones to hit Fiji in recent history include Cyclone Winston (2016), the strongest cyclone ever recorded in the Southern Hemisphere, Cyclone Yasa (2020) which was the second-strongest cyclone on record to make landfall in Fiji, and Cyclone Harold (2020). For these three most recent storms, international support was mobilized, including financial support from the Central Emergency Response Fund (CERF).

Climate change leading to warmer oceans is having a significant impact on tropical cyclones, making them more frequent, intense, and destructive. In addition, climate change is also expected to lead to stronger storm surges, more extreme rainfall events, and rising sea levels. Fiji's government estimates (NDC) that by 2050, over 6.5 per cent of Fiji's GDP could be lost due to tropical cyclones and floods, with the number of people being pushed into poverty every year increasing by 25 per cent. While the Government of Fiji has made significant investments in disaster preparedness and resilience, the country remains at high risk from the loss and damage caused by tropical cyclones.

Fiji's National Adaptation Plan (NAP)² and National DRR policy³ identifies tropical cyclones as a major climate hazard that the country is facing and will continue to face in the future. The NAP outlines several measures that Fiji can take to address the risks posed by tropical cyclones, including the building of community resilience, as well as strengthening of early warning systems and disaster response preparedness.

Anticipatory action

By using early warning systems and scientific advances in predicting disasters, anticipatory action maximizes the window of opportunity between the moment of prediction and the arrival of a specific forecasted shock to trigger interventions that prevent or mitigate imminent humanitarian impacts.

Anticipatory action is defined as acting ahead of predicted hazards to prevent or reduce acute humanitarian impacts before they fully unfold. Anticipatory action works best if triggers and decision-making rules (the model) and activities (the delivery) are pre-agreed to guarantee the fast release of pre-arranged financing (the money).

In line with best practices for anticipatory action, this framework pre-agrees the tropical cyclone forecasting trigger and the activities which are launched thanks to pre-arranged financing. In addition, documenting evidence and learning is part of the framework.

² https://www4.unfccc.int/sites/NAPC/Documents/Parties/National%20Adaptation%20Plan_Fiji.pdf

³ <https://www.ndmo.gov.fj/ndrrp/>

2. The Model: Forecasts, triggers, and activation protocol

Model/forecast

Under the World Meteorological Organization's Tropical Cyclone Programme, the Regional Specialized Meteorological Centre (RSMC) in Nadi has the responsibility for detecting and forecasting tropical cyclones (TCs) in its zone of responsibility, which is from the equator to 25°S and from 160°E to 120°W (spanning the international date line)⁴. As such, RSMC Nadi's TC forecasts are the internationally recognized authoritative source for TCs in this zone. RSMC Nadi is operated by Fiji Meteorological Services (FMS).

RSMC Nadi issues official TC track forecasts with a 72-hour lead time, which include the TC's forecasted category and position, as well as 5-day TC formation risk outlooks. This framework uses the official 72-hour track forecast for its action trigger. To provide more advance warning for implementation, this framework also includes a readiness trigger, which is based on RSMC Nadi's unofficial 120-hour forecast (an additional 48 hours)⁵.

Trigger Conditions

This framework will be triggered by a tropical cyclone that is forecasted by RSMC Nadi to be a:

- **Category⁶ 4 or 5 while within 250 km of Fiji, or**
- **Category 3, 4, or 5 while making landfall in Fiji.**

Every time RSMC Nadi produces a tropical cyclone track forecast, it will be evaluated to determine whether the trigger conditions have been met.

The framework follows a 2-stage trigger activation, each part of which is triggered based on the lead time of the points in the forecast:

1. **Readiness trigger:** If any point in the 120-hour forecast meets the trigger conditions, the readiness trigger is met.
2. **Action trigger:** If any point in the forecast up to a lead time of 72 hours meets the trigger conditions, the activation trigger is met.

By definition, the action trigger has a detection rate of 100% (*i.e.*, any cyclone whose actual track meets the trigger conditions will cause the framework to activate). This is because the actual track of the cyclone is effectively included in the forecast, as the forecasted position at a lead time of 0 hours.

Trigger Monitoring Process

Every tropical cyclone track forecast produced by RSMC Nadi is evaluated to determine whether the trigger conditions have been met. The process involves the following steps:

1. A 120-hour forecast is produced by the Fiji Meteorological Services Senior Scientific Officer on duty on the Tropical Cyclone Forecasting Bench.
2. The Officer emails the forecast, in its standardized CSV export format from the TC Module software to the Data Science team at the OCHA Centre for Humanitarian Data.⁷
3. An automated script⁸ evaluates whether the trigger conditions have been met for either the readiness trigger or the activation trigger.

⁴ <https://community.wmo.int/en/tropical-cyclone-regional-bodies>

⁵ Due to the relative uncertainty of the additional 48 hours of forecast, this part of the forecast is used only to determine whether the readiness trigger has been met.

⁶ The Category in the Australian scale, corresponding to 10-minute sustained wind speeds of >107 kn for Cat 5, 86-107 kn for Cat 4, and 64-85 kn for Cat 3.

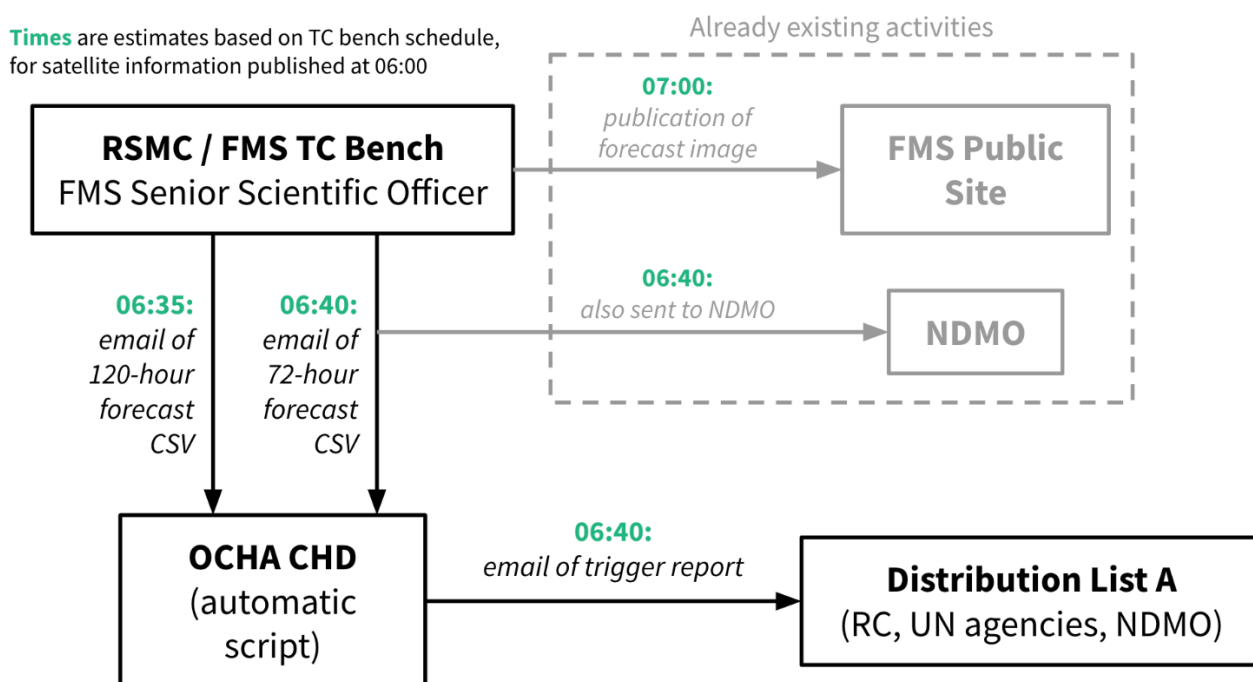
⁷ data.science@humdata.org

⁸ All code used for analysis in this framework is available publicly at <https://github.com/OCHA-DAP/pa-aa-fiji-storms>

4. The automated script sends an email to the Distribution List defined in the Annex to notify that the trigger has been met.

As a backup to the above steps, the Data Science team email will also be included on emails of the 72-hour official forecast from RSMC Nadi to the NDMO, which happen after the 120-hour has been truncated to 72 hours (an already existing activity, undertaken for the NDMO to produce impact forecasts).

This process is outlined in the figure below.

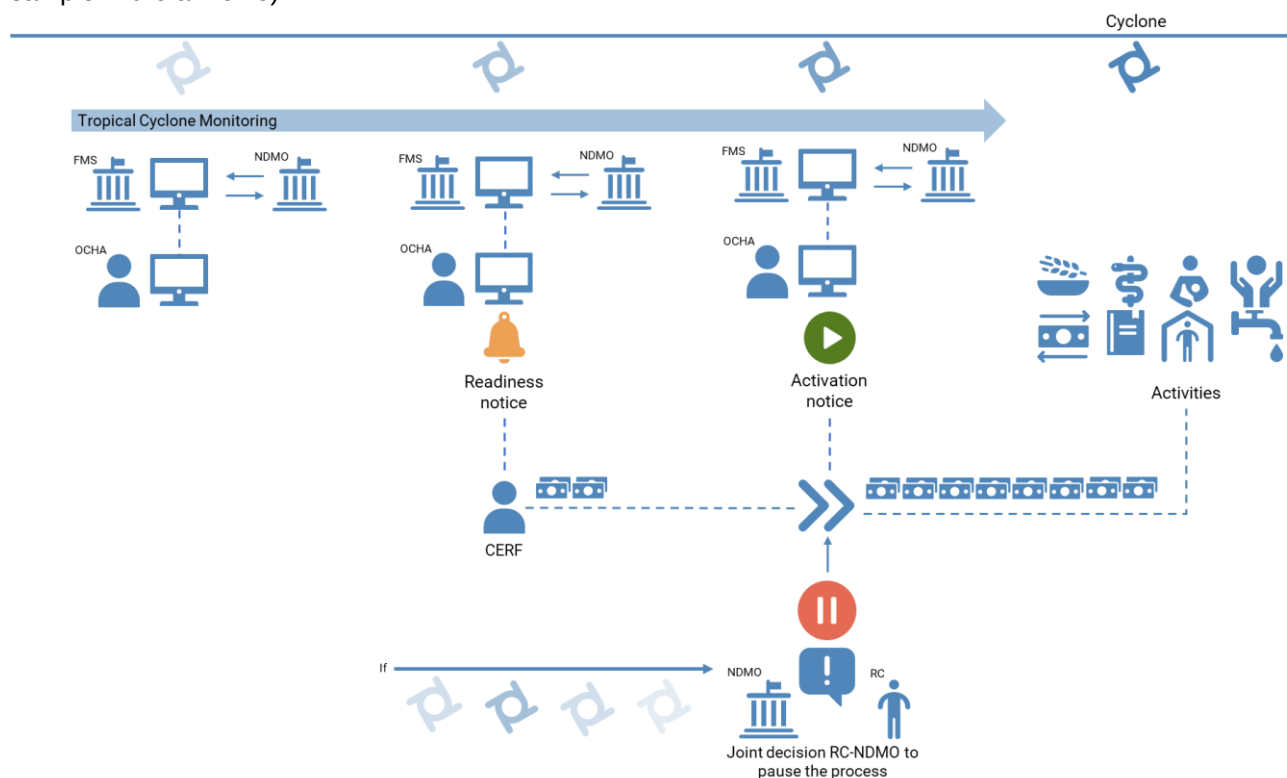


Activation protocol

Who	What	How	When
RSMC Nadi / Fiji Meteorological Services	Send 120-hour forecast to OCHA Centre for Humanitarian Data	Email of CSV to data.science@humdata.org	As soon as forecast is produced, and before forecast is truncated to 72 hours
OCHA Centre for Humanitarian Data	Evaluate whether trigger has been met, and send email to distribution list A	Using automated script	Automatically as soon as forecast CSV is received from RSMC Nadi
NDMO	Share the trigger report to all members of national clusters involved	Email by NDMO to distribution list B (Annex 3)	As soon as notification of trigger is received
CERF	When a readiness trigger is issued (or an action trigger without a previous readiness trigger), CERF will immediately send the approval letters to the agencies	Endorsed framework and CERF application by RC and Emergency Relief Coordinator (ERC) Pre-approved CERF approval letters to UN agencies	As triggers are reached
RC	The RC can pause the activation process upon discussion with the NDMO	Direct contact between RC and NDMO (it can be facilitated by OCHA HoO) followed by email notification to distribution list A (Annex 3)	Once the framework is activated, in the possible outlier cases where a track of a storm clearly would result in minimal impact.
NDMO	The NDMO can discuss with the RC if it is the case to pause the activation	Direct contact between RC and NDMO (it can be facilitated by OCHA HoO)	Once the framework is activated, in the possible outlier cases where a track of a

		followed by email notification to distribution list B (Annex 3)	storm clearly would result in minimal impact.
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To support geographic prioritization, OCHA CHD will produce an information email on the cyclone track (email sample in the annex 5).



Any triggering of the framework should lead to action. However, for theoretically possible outlier cases where a track of a storm clearly would result in minimal impact, the Resident Coordinator, in consultation with the government, may pause any activation.

Challenges and potential improvements in the future

- Need for impact-based forecasting after triggers are reached. This could start already but be improved continuously.
- Leadtime for action may be extremely short and thus more investment is needed in the science of cyclone forecasting to increase lead-times and accuracy.
- Tropical cyclones through the lens of windspeed is only one hazard resulting from storms, alongside among other things rainfall and storm surge. Fiji faces other predictable hazards which are not included in this anticipatory action framework. In the future, it was agreed to explore further how rainfall could figure into a potential improved framework, while anticipatory action for other hazards may also be explored.

3. The Activities: Activities to mitigate the impact of tropical cyclones

Overview

Tropical cyclones have a profound, devastating impact on communities in Fiji experiencing greatest vulnerability. These populations, often living in precarious housing and lacking essential resources, bear the brunt of disasters. While they possess the knowledge and skills to mitigate the impacts of cyclones, they often lack the necessary materials and resources to effectively reduce their vulnerability to them.

There is only a very short window of opportunity between a trigger and impact of tropical cyclones in Fiji. In addition, some areas are remote and difficult to access, especially with impending storms. This combination may create challenges to carry out anticipatory action in full before the impact of the cyclone. This is why the partners have resolved to act as fast as possible and as anticipatory as possible once a trigger is reached. However, in some circumstances, especially for remote areas, assistance may only reach communities after the cyclone has passed, turning ex ante anticipatory action into ex-post early action.

Selection Criteria

Anticipatory actions aim to mitigate the impacts of out-of-the ordinary tropical cyclones by targeting those people who face the greatest vulnerability. The following criteria were applied to select the pre-agreed activities:

1. Anticipatory character: Is the action effective in preventing or reducing the humanitarian impact of the shock?
2. Ability to deliver within the window of opportunity: Is it possible to carry out the action effectively with the available forecast lead time, i.e. in the window of opportunity?
3. Operational capacity: Do the agency and its IPs have the institutional capacity (thematic, logistic, administrative, financial, human resources) to implement the action effectively given the lead time and scale?
4. No regrets approach: In the case of a false alarm, will the proposed actions benefit rather than negatively affecting the targeted population?

Activities⁹

Activity 1. Multi-Purpose Cash co-lead by Ministry of Women, Children and Social Protection (MWCSP) and WFP – Safety and Protection Cluster

Objective: Mitigate the impact of the cyclones on household food and nutrition security and general wellbeing of their families.

Description: Given the short lead time, unconditional multi-purpose cash transfer (MPC) is a major component of the framework. WFP, in partnership with the Ministry of Women, Children and Social Protection (MWCSP), will support a projected 15,000 households (75,000 people) with anticipatory cash transfers in areas identified to be in tropical cyclones path at the time of the activation trigger. A transfer value of FJD200 endorsed by the Fiji cash working group will be transferred by the DSP to targeted households. In the absence of minimum expenditure basket, transfer value determination was based on the default transfer value and aligned to Fiji basic needs poverty line.

WFP will channel resources through the government for provision of cash assistance to the existing beneficiaries of social support schemes identified by the Government under four parts of the national social protection scheme namely: 1) the Family Assistance Scheme, 2) the Care and Protection Allowance, 3) the Disability Allowance and 4) the Social Pension Scheme. This MPC are designed to enable targeted households to prepare for the cyclone (evacuation costs, protecting livestock and homes etc.), buy essential needs (food, medicines etc.) and mitigate the impact of the cyclones on household food and nutrition security and general wellbeing of their families. WFP will collaborate with government and UN agencies in development of early warning messages under the leadership of NDMO and support the Department of Social Protection (DSP) to disseminate early warning messages including cash assistance messages to the final list

⁹ For a detailed description of UN activities see the relevant CERF projects at <https://cerf.un.org/>. There is also a summary table in the annex.

of targeted households. The DSP will engage field-based staff to support community sensitization about the tropical cyclones and potential for assistance before impact. Messages about assistance will be designed and delivered in such a way they do not raise expectations for cash transfers in areas not affected by the tropical cyclone.

Partners: Ministry of Women, Children and Social Protection and Ministry of Finance

Challenges and potential improvements in the future:

Sudden onset disasters, such as tropical cyclones, affect everyone in their path, whereas the use of a social registry for targeting restricts anticipatory support, such as multipurpose cash, to households in the social protection beneficiary registry. In addition, while the use of the national social protection beneficiary registry contributes to the strategic objective of strengthening government systems, there is limited oversight over inclusion and exclusion errors when a disaster occurs. As a mitigating measure, regular deduplication, graduation, and retargeting to include households below the poverty line can increase the likelihood that the most vulnerable individuals will be included on the beneficiary list. While the above is a medium to long-term solution, there is a clear need for early action plans, particularly when using the government beneficiary registry. The early actions can be activated promptly after the disaster has passed to support and prioritize those who were affected but not targeted through government beneficiary registry (horizontal expansion). The AA pilot is an opportunity to showcase how existing government policies and legislative frameworks for financing disaster risk management, social protection, and climate change adaptation can reduce impact of climate disaster through anticipatory actions. This strategy ensures that AA interventions are localized, owned by the government and hence sustainable and cost-effective.

Activity 2. Fishing-boat strengthening kits and water-tight drums co-lead by Ministry of Agriculture and FAO/Fijian Red Cross

Objective: Reducing the loss of livelihoods by securing boats, tools, and cash crops.

Description: To mitigate the potential impacts on the agriculture sector, FAO and the Fiji Red Cross, in agreement with the Ministry of Agriculture, will distribute fishing-boat strengthening kits to at-risk fisherfolk and water-tight drums to farmers. The fishing-boat strengthening kits will enable fisherfolk to pull their boats ashore and store them in secure areas such as mangroves or along sandbars, where they can be tightly secured. Fisherfolk will also be presented with the opportunity to store their gear in the water-tight drums. Water-tight drums present a viable solution for farmers to securely store their cash crops, providing a dry sanctuary that mitigates the risk of damage. It has dual purpose: firstly, to facilitate the preservation of crops, and secondly, to provide a repository for clippings. By pre-emptively stockpiling crops prior to the cyclone's arrival, the potential for crop damage and loss can be mitigated. The decision to use the drum for either crops/fishing gear, or both, will be at the discretion of the household. This will be complemented by early warning messages delivered to high-risk households in accordance with the established communication protocols outlined by NDMO and other UN agencies.

Partners: Ministry of Agriculture, Fiji Red Cross and IFRC

Challenges and potential improvements in the future: One of the primary obstacles is related to logistics. It will be crucial to preposition and store items in advance of the season to maximize their impact. While the necessary items can be sourced locally, they are predominantly found in Suva or Nadi. Ensuring their accessibility from various locations across the country and maintaining their safety in readily accessible warehouses will be of utmost importance. Simulating and practicing the delivery arm of these actions will be critical in the 'off season' to address any unforeseen challenges with getting the goods from the warehouses to at-risk families. As we progress, we'll need to refine and scale up this process, envisioning what the expansion might entail. This will require collaboration with the Ministry of Agriculture, building on the lessons learned during this initial pilot stage.

Activity 3. Maternal and reproductive health supplies and GBV protection co-lead by Ministry of Health and Medical Services (MHMD), MWCSP and UNFPA

Objective: Safeguard the health and well-being of women and girls to be affected by impending cyclones

Description: Partnering closely with the Ministry of Health and Medical Services (MHMS) and MWCSP, as well as local NGOs and associations, UNFPA will provide essential maternal and reproductive health supplies, alongside targeted GBV protections, to women and girls in forecasted cyclone impact areas. UNFPA's intervention has a strategic emphasis on minimizing life-threatening risks associated with Sexual and Reproductive Health (SRH) and Gender-Based Violence (GBV). These efforts encompass the distribution of Pacific-customized kits, complemented by informational messages stressing the importance of seeking timely support and care from trained healthcare professionals and other service providers. These kits comprise the following: clean delivery kits for pregnant women in the community, dignity kits for women aged 15-49, including those with disabilities, and menstrual hygiene kits for girls aged 10 and above. This comprehensive strategy aims to safeguard the health and well-being of women and girls in the face of impending cyclones, addressing their unique needs and vulnerabilities.

Partners: Ministry of Health and Medical Services, Ministry of Women, Children, and Social Protection, Medical Services Pacific, Fiji Disabled People's Federation, and Reproductive and Family Health Association of Fiji.

Challenges and potential improvements in the future:

Data Accuracy - Accurate and timely data is essential for successful anticipatory action, especially for vulnerable populations in crisis situations. Inaccurate data can lead to the misallocation of resources. UNFPA will potentially try to address this challenge by:

- Utilizing data analytics and modelling techniques such as the MISP calculator to predict the needs of vulnerable populations based on historical data and trends.
- Engaging with local communities to establish nature-based monitoring mechanisms that provide real-time data on emerging risks.

Logistical Constraints - Delivering supplies and implementing anticipatory measures in remote and at-risk areas can be challenging, especially when time is of the essence. To overcome logistical constraints, UNFPA will consider:

- Identifying strategic locations to pre-position essential supplies, ready for rapid deployment when an emergency is anticipated.
- Building partnerships with local organizations, providing them with the resources and training necessary to assist in anticipatory actions.

Coordination and Collaboration - Coordinating anticipatory action efforts with various stakeholders, including government agencies, UN agencies, and community leaders, is vital for a successful response but can also be complex and challenging. UNFPA will work towards enhancing coordination by:

- Advocating for collaboration on emergency preparedness activities and information sharing on challenges and good practices in the health cluster and protection cluster.
- Conducting joint planning and simulation exercises with relevant partners to improve coordination for AA activities.
- Engaging with local communities to build their capacity in disaster preparedness and response, fostering a sense of ownership and participation.

Activity 4. Distribution of WASH kits co-lead by the Ministry of Health and Medical Services (MHMS) with UNICEF

Objective: Mitigate the impact of the cyclones on access to water and sanitation services and ensuring effective hygiene practices.

Description: UNICEF will work with the MHMS to improve and maintain access to Water, Sanitation, and Hygiene (WASH), through the following activities:

- Coordinate the WASH cluster response with technical advice, such as assessments, (inter-)cluster coordination, information management (4Ws) and reporting.
- Prepositioning of WASH related non-food items at likely impacted evacuation centres and key public infrastructures (hygiene kits, water containers, menstrual hygiene management materials).
- Support coordination and innovation management across government led WASH, cluster.
- Targeted support at evacuation centres to ensure WASH response is gender and disability sensitive.
- UNICEF will monitor and evaluate the implementation of the project to measure the impact of anticipatory action in cooperation with the Government counterparts.

Partners: Ministry of Health & Medical Services, Ministry of Education and Human Resource Development, WASH Cluster CSO partners (Contingency partnerships in progress).

Challenges and potential improvements in the future:

Much work needs to be done to improve preparedness in Fiji including: training for cluster coordinators and information managers, onboarding of contingency partnerships for rapid response, increasing number of long-term agreements with contractors to deliver local supplies, review and strengthen local long-term agreements with NFI suppliers. These are all ongoing actions but as cyclone season arrives are still in progress to develop and may not fully be in place if there is an early activation.

Activity 5. Learning materials co-lead by the Ministry of Education and UNICEF

Objective: Ensure that children return to safe learning environments and are provided with relevant learning resources to continue their education.

Description: To provide vulnerable children access to a secure learning environment, UNICEF will distribute school-in-a box kit to targeted schools. UNICEF and the Ministry of Education (MoE) will also provide vulnerable children and their care in with access to PFA (Psychosocial First Aid).

- Coordinate the Education cluster response with technical advice, such as assessments, (inter-)cluster coordination, information management (4Ws) and reporting.
- Prepositioning of Education related non-food items at schools (tents, school kits).
- Support coordination and innovation management across government led Education cluster.
- UNICEF will monitor and evaluate the implementation of the project to measure the impact of anticipatory action in cooperation with the Government counterparts.

Partners: Ministry of Education and Human Resource Development (MEHRD), Education Cluster CSO partners.

Challenges and potential improvements in the future:

- Detailed data on school populations and vulnerability not available. Better data in the future can lead to better targeting of supplies and improved prepositioning.
- Tents need to be sourced internationally and lead times for replacement can be slow.
- Local expertise and capacity to erect tents without training is limited. Further training will be needed in the future to support first responders.

Activity 6. Shelter strengthening kits co-lead by Ministry of Rural & Maritime Development and Disaster Management and IOM with technical support from IFRC

Objective: To minimize the damage caused by cyclones on weak shelters belonging to vulnerable community members.

Description: IOM, with technical and coordination support through Fiji's Shelter cluster and IFRC, will work with active Shelter Cluster members, agencies, offices, and organizations to provide the following:

- IEC materials on shelter reinforcement and the basics of tying down a shelter, as well as on safe evacuation and local service providers in the event of an emergency.
- Ropes, fasteners, screws, tarpaulins, and other shelter materials to support the strengthening of shelters ahead of the landfall of a cyclone.
- Construction Tools to support communities in the shelter strengthening process.

Partners: Ministry of Rural & Maritime Development and Disaster Management, IOM, IFRC, Fiji Red Cross, Habitat for Humanity

Challenges and potential improvements in the future:

There are several envisaged challenges to shelter interventions conducted within the AA Framework, the majority of which revolve around the provision of support to the most vulnerable or in need communities. Three core challenges have been discussed amongst the National Shelter Cluster members: 1) with shelter materials often spiking in cost after a cyclone makes landfall, deploying prepositioned kits before landfall leaves a risk of supporting households who may be less in need than others following landfall. With the spike in price and less prepositioned kits, response efforts may be hampered. 2) The timeframe of the AA activities leaves little room for capacity building, so interventions will need to largely rely on construction knowledge already present in any given community. The need to manage expectations for what can be achieved in this window, as well as focusing targeting on communities with experience in repairing/strengthening shelters, is a challenge. 3) Reaching maritime islands will pose a significant challenge to the cluster as most warehouses and stocks are in Viti Levu and Vanua Levu, with the greatest concentration in Suva and Nadi. Movement to maritime islands right before a cyclone may be too high of a risk.

Activity 7. Safety and Protection horizontal support co-lead by the Ministry of Women, Children and Social Protection (MWCSP) and UN Women

Objective: To ensure safety and protection concerns are assessed and appropriately factored into post needs assessments and responses of the different key sectors.

Description: From the alert or trigger phase, social welfare officers and women's interest officers in the rural and maritime areas, as part of their preparedness activities to respond, are usually already moving through location where their services are based to check on the facilities, the services it's meant to provide in safety and protection. During the past year, the cluster had also revised their rapid assessment form to be deployed on Kobo. A series of workshops has been held in the various divisions over the past year to finalize the form. During the readiness phase, the officers will be refreshed on the use of the form on Kobo and how to collect protection data. This will be facilitated online whilst in their various rural locations.

Partners: Department of Women & Department of Child services and Social Protection and Fiji Safety and Protection Cluster Members.

Challenges and potential improvements in the future:

- The designation of staff (usually public servants) that would support protection assessments will only be determined at the time of the looming emergency. In future, the ministry, as part of their preparedness could already obtain a list of individual staff that would be on call and able to deploy for assessments as soon as restriction to movements has been lifted.
- Virtual training will only depend on the strength of the network in the divisions and the available devices which assessors will use for accessing Kobo and collecting data. In future, a series of data collection devices could be acquired and issued out on receipt of a warning (as an asset of the organization) so that there won't be any issues in accessing the kobo form. Data credit can also be pushed out to the sim card on the devices to ensure connectivity during the refresher trainings.

Activity 8. Health and Nutrition Cluster Coordination co-lead by the Ministry of Health and WHO

Objective: To work in support of the Ministry of Health and other national authorities to improve the health and nutrition outcomes of affected populations through timely, predictable, appropriate and effective coordination of health and nutrition action in emergencies.

Description:

- Support Fiji's Health and Nutrition Cluster in coordinating national-level preparedness, emergency response and recovery efforts, as requested/directed by MOH leadership.
- Support Fiji's Health and Nutrition Cluster with assessments, preparedness and service delivery in emergencies through the provision of technical support, operational support (i.e. implementation), and in-kind support, as requested/directed by MOH leadership.
- Support Fiji's Health and Nutrition Cluster to identify core advocacy concerns, including resource requirements, and contribute key messages to advocacy outlets.

Partners: Fiji Ministry of Health and Medical Services, Fiji Health and Nutrition Cluster, Adventist Development and Relief Agency (ADRA), Empower Pacific, International Federation of Red Cross (IFRC), International Planned Parenthood Federation (IPPF), The Pacific Community (SPC), Save the Children, United National Population Fund (UNFPA), United National Children's Fund (UNICEF), World Health Organization (WHO).

Challenges and potential improvements in the future: A resilient health system must include the ability to both maintain the provision of basic health care and respond to health system shocks. A strong and equitable health system is resilient in the face of potential threats, including but not limited to the climate crisis, public health emergencies and disasters. The ongoing work of health and nutrition cluster partners to build a resilient health system will support Fiji throughout the preparedness, anticipatory action, response and recovery phases.

Activity 9. Common approach to Early Warning and Community Sensitization facilitated by UNICEF and UNDRR

Objective: Protect and minimize the loss of lives and livelihoods by disseminating timely and actionable early warning messages to communities via multiple channels.

Description: Building on the existing "Get Ready Disaster Happen" multi-media campaign, this activity will support the development and dissemination of timely early warning messages to the public through radio, television, social media and SMS platforms. The early warning messages are linked to the specific readiness and activation triggers and provide sector specific messages, including WASH, Health, Nutrition and Protection issues, Food Security, and education. These messages will be broadcasted in English, iTaukei and Hindi languages. During the readiness trigger, the messages will constantly air on radio and television platforms, through SMS blasts and social media. The frequency of these messages on all platforms will increase once the Action Trigger is Activated. Additionally, through the partnerships with CSO partners, volunteers will be mobilized into communities that would fall on the potential path of the disaster to assist in cyclone preparations. Turaga ni Koros, village leaders and advisory counsellors who would be sensitized on Anticipatory Action will be engaged through digital platforms and mobilized in their respective communities.

Partners: Ministry of Rural & Maritime Development and Disaster Management; National Disaster Management Office (NDMO); Communication Cluster (CCE WG), United Nations Office for Disaster Risk Reduction (UNDRR), and World Health Organization (WHO).

Challenges and potential improvements in the future: Potential Challenges and potential improvements in the future are:

- Dissemination of messages through media and SMS platforms can be disrupted should the telecommunications infrastructure be damaged by the disaster.
- Community mobilization can be hindered with restricted movements in communities.
- Need to revise and develop more messages with the input from each cluster/sector.

Logistics

Efficient operational logistics are paramount for the success of Anticipatory Action in Fiji. UN Agencies in Fiji have access to the UNCT SharePoint, a central platform for operational collaboration and information exchange. To streamline and improve operational logistics, each UN Agency's program and supply units have access to the Long-Term Agreements (LTAs) Directory, which is available on the UNCT SharePoint. This directory contains comprehensive information and instructions for effectively piggybacking existing LTAs, thereby leveraging the global UN Mutual Recognition Agreement. It is noteworthy that all UN agencies involved in the current proposal have endorsed a Mutual Recognition Statement. It's crucial to emphasize that

OMT Representatives and Operations Teams in all UN Agencies across the Pacific region are well-versed in the operational procedures associated with this Mutual Recognition Agreement.

Access to LTA Directory: Within the UNCT SharePoint, all UN Agencies in Fiji can access the Long-Term Agreements (LTA) Directory. This directory serves as a valuable resource, providing comprehensive information and instructions on how to efficiently piggyback existing LTAs. The utilisation of this directory is instrumental in capitalising on the global UN Mutual Recognition Agreement.

Mutual Recognition Statement: All UN agencies involved in the current proposal have signed a Mutual Recognition Statement, emphasising the commitment to cooperation and harmonisation in operational logistics. This shared agreement fosters a collaborative and unified approach among UN Agencies, ensuring streamlined and efficient logistics processes.

Global LTA Requests: In addition to the UNCT SharePoint, UN Agencies in Fiji have the flexibility to request copies of global LTAs listed on the United Nations Global Marketplace (UNGM) website. This option allows each UN Agency to directly contact the respective UN Agency responsible for a particular LTA, ensuring timely access to essential resources.

Piggybacking Process: The well-established piggybacking process is an efficient means of streamlining operational procurement and logistics. This process extends to various types of LTAs, including those related to transport, non-food items (NFIs), warehouse storage capacity, and more. UN Agencies in Fiji can leverage this process to bolster their operational capabilities.

UN Agencies in Fiji, through their access to the UNCT SharePoint and commitment to the Mutual Recognition Agreement, possess the necessary tools and cooperative spirit to optimize logistics and procurement and distribution processes. Leveraging global LTAs from UNGM further enhances their operational capabilities.

Challenges and potential improvements in the future

- Limited timeframe for preparatory work ahead of 2023 cyclone season and limited availability of resources. This meant the pilot had to focus on a prioritized set of interventions which are feasible and practical. This may also compromise the collection of baseline data. Limited time and resources hinder comprehensive coverage of vulnerable populations. Furthermore, coordination among government systems, UN Agencies and other partners may require additional efforts to align strategies.
- A common approach to targeting is being explored. Issues such as data sharing and limited time are hindering its scope for the 2023 cyclone season.
- Common logistics is being explored under the leadership of UNICEF and the national Logistics cluster. Limited time for planning might not make it available for the 2023 cyclone season.
- Remote areas such as the Eastern division are difficult and impossible to reach with assistance on the first forecast. Geographic barriers may delay or impede supply distribution to remote areas.
- TCs' unpredictable paths make targeting challenging, requiring a balance between certainty and impact. Establishing effective vulnerability criteria is thus crucial.
- Obtaining precise and current data on at risk populations, especially in crisis situations, presents challenges. Inaccurate data can result in the misallocation of resources. For instance, whilst WFP's use of the national social protection beneficiary registry supports the strategic outcome of strengthening government systems, there is limited control on the inclusion and exclusion errors when the disaster strikes. An impact evaluation of the AA transfers can generate lessons for the Department of Social Protection on how to improve beneficiary targeting when using government systems by understanding impact of targeting from existing beneficiary registry on inclusion and exclusion errors. This learning would lead to enhancement of beneficiary registry.
- There are restrictions in accessing highly at-risk groups in peri-urban areas.
- Involving local partners and engaging local communities in preparedness, targeting, and activities is a challenge.
- Better integration with preparedness efforts by development organizations is needed. This includes potential involvement of regional organizations such as the South Pacific Community.

- Exploring collaborations with the private sector through initiatives like the Fiji Disaster Business Resilience Council and the OCHA and UNDP-led Connecting Business Initiative (CBI).
- There is a clear need to have timely response plans along with anticipatory actions plans, especially when using the government beneficiary registry. The timely response actions can be activated immediately after the disaster has passed to support those that have been impacted but were not supported by the AA pilot.
- Additional resources would support capacity building. For instance, investing in capacity-building initiatives for local healthcare providers, community leaders, and volunteers would enhance their effectiveness in responding to SRH and GBV needs during emergencies.
- The CERF-Anticipatory Action Framework will ensure that the human rights and protection needs of specific groups of population that are often left behind -such as Persons with Disabilities (PWDs), Persons with Albinism, women, children, elderly, and ethnic or other minorities are equally protected by the CERF-Anticipatory Actions interventions in Fiji.¹⁰ In order to comply with these objectives, OHCHR Pacific Regional Office has proposed to collaborate with the CERF-Anticipatory Action in Fiji to provide technical cooperation and capacity building for the national clusters and for the NDMO on the integration of the HRBA and SDGs in the context of emergency responses to natural disasters.

¹⁰ Fiji ratified the nine United Nations (UN) core human rights treaties. Through the ratification of the international human rights treaties, Fiji assumed international legal obligations to always respect, protect, and fulfill human rights, including during the state humanitarian responses to natural disasters. Additionally, Fiji committed to work in line with the United Nations 2030 Agenda for sustainable development and towards the achievement of the Sustainable Development Goals (SDGs). Taking into consideration both frameworks, it becomes essential to integrate the Human Rights Based Approach (HRBA) values and the principle of “leaving no one behind” in the humanitarian responses to natural disasters, including to respond to tropical cyclones.

4. The Money: Pre-arranged financing

Pre-arranged finance to fuel anticipatory action is critical, especially in sudden onset disasters, to enable timely, effective anticipatory action. A key recommendation from the USG/ERC was to find co-financing for coordinated anticipatory action in Fiji to ensure available funding is mutually reinforcing. Of course, partner agencies invest considerable resources in building coordinated anticipatory action, but also for preparedness already.

This framework for anticipatory action is not limited to CERF-funded activities. There are various other pooled funds, bilateral donors and UN agencies' internal reserve funds that can, within their own established criteria and in complementarity, finance part of the collective approach when the need arises. Currently, WFP, FAO, UNICEF, UNFPA and UNW have been able to secure additional funds.

Financing overview

Funding overview in US\$ for coordinated framework

	CERF	Additional agency financing	Total
WFP	948,990	500,000	1,448,990
FAO	274,990	200,000	474,990
UNICEF	774,187	400,000	1,174,187
UNFPA	200,075	200,000	400,075
IOM	200,000		200,000
UNW		50,000	50,000
WHO		30,000	30,000
Others			
Total	2,398,242	1,380,000	3,778,242

Central Emergency Response Fund (CERF)

The Emergency Relief Coordinator agreed to allocate up to \$2.4 million from CERF to support anticipatory action in Fiji. CERF will disburse funds to anticipatory action frameworks on a no-regrets basis once three conditions are met:

- Endorsement by the relevant Resident/Humanitarian Coordinator of the Anticipatory Action framework (this document) and the CERF application package comprising an application chapeau, agency-specific project proposals and agency-specific budgets (in annex);
- Pre-approval by the Emergency Relief Coordinator of the Anticipatory Action framework (this document) and the CERF application package comprising an application chapeau, agency-specific project proposals and agency-specific budgets (in annex); and
- Activation of either the readiness trigger or the action trigger within a maximum of two years from the ERC's pre-approval of the framework.

CERF disbursement is guaranteed once a trigger is reached. As the CERF funding will be activated and distributed as automatically as possible immediately once the defined trigger(s) is/are reached. CERF will disburse funds on a no-regrets basis. While a portion of the fund can be used immediately upon disbursement for readiness activities, the remaining portion of the funding can only be used if and when an action trigger is reached. The amount of funding for readiness vs action activities will be clearly defined in the pre-approved project proposals and budget. The need to distinguish between these two cost categories in advance is important given the different potential scenarios.

Pre-arrangement and/or receipt of CERF funding for anticipatory action does not preclude or guarantee additional CERF funding for a complementary rapid response to save lives resulting from the same or other

shocks. This means that the pre-arranged funding by CERF for anticipatory action is for the hazards and triggers outlined in this framework. Should there be additional needs or other disasters, CERF's other funding modalities can be requested.

Exact details on financing per agency can be found in the CERF application documents annexed.

Co-Financing

With support from the German Federal Foreign Office, a sum of USD500,000 has been allocated by WFP to be disbursed upon confirmation of activation triggers to co-finance the CERF allocation intended to reach 10,000 households with multipurpose cash assistance, as well as other readiness and activation costs. Five thousand (5000) additional households will benefit from WFP's co-financing. In total 15,000 households are planned to receive multipurpose financial assistance through the Anticipatory Action pilot project

Under FAO's Anticipatory Action Window, with generous support from the German Federal Foreign Office, a sum of USD 200,000 has been strategically allocated to Fiji. This allocation is in reaction to the elevated probability of an above-average cyclone season in the region. The funds will be directed towards facilitating anticipatory action activities by prepositioning and distributing watertight drums and reinforcing fishing boats. Furthermore, this financial support plays a pivotal role in fostering collaboration with the Fiji Red Cross, thereby streamlining the logistical and operational procedures to expedite action in light of the trigger.

UNICEF is strategically poised to proactively address humanitarian needs in target communities, with a dedicated allocation of USD 400,000 from the Global Humanitarian Thematic Funds. This funding enables UNICEF, both independently and in collaboration with its trusted partners on the ground, to swiftly provide vital assistance to those most vulnerable in the face of impending crises. Furthermore, UNICEF's ongoing two-year work plan in cooperation with the Government of Fiji demonstrates the organization's commitment to preparedness and anticipatory action. Within this comprehensive plan, there exists a flexibility that allows for the seamless transition of existing activities into anticipatory actions when required. This proactive approach underscores UNICEF's dedication to ensuring the well-being and safety of communities at risk, reinforcing its role as a steadfast humanitarian leader.

The proposed AA funding of 212,413 USD will complement UNFPA's ongoing disaster preparedness and response activities (200,000 USD) in Fiji supported by donors such as USAID-BHA and DFAT. Both UNFPA core resources and donor resources will be used to reach additional beneficiaries of the AA pilot initiative.

WHO's co-financing supports work in Anticipatory Action on cross-cutting issues including contracting a company to test early warning messages and technical support to Fiji's Health and Nutrition Cluster

UNW's co-financing is an extension of the work under Information Management that is already supporting the MWCSF on. These funds are from the Core Humanitarian Funds of the organization.

Challenges and potential improvements in the future

- While financing is increasingly available to carry out anticipatory actions, collective anticipatory action requires an investment in building the framework. These "start-up costs" (or "build costs" as part of a "build – fuel – evaluate" cycle of frameworks) are incurred by participating agencies prior to the activation of a framework set up effective anticipatory actions. These costs may also include projects which enable a collective approach (e.g. common targeting approaches) and other costs that pooled funds, including the CERF, traditionally do not fund.
- To enable a timely response within a short window of opportunity in a geographically very diverse country such as Fiji, anticipatory action at scale requires commensurate disaster risk preparedness, including the prepositioning of goods. There is limited funding available for such emergency response preparedness, thus limiting the scope of this anticipatory action framework.
- It must be understood that anticipatory action may support some of the most vulnerable communities facing the impact of a tropical cyclone. Yet, the storms are still likely to cause damage beyond the maximum number of people targeted, beyond the geographic reach or the scope of sectors covered by this anticipatory action framework. More thought needs to be invested in ensuring linkages between

emergency response preparedness, anticipatory action, rapid response and early recovery, including around community engagement, activities and financing.

5. The Learning

Any activation of this framework will offer enhanced learning opportunities on implementing anticipatory action and further provide evidence to support scale-up of the approach in the future. Evidence generation and learning are a critical key to adjustment and scale up of the approach with government and non-government stakeholders. To support this process, and project learning, monitoring and evaluation committee has been convened by OCHA with partners to this framework to coordinate learning and evidence gathering on anticipatory action. One key aspect of learning will be to assess the impact of the framework against the premise of where possible, anticipatory actions may lead to a faster, more efficient, and more dignified approach which also may protect development gains. Another key learning will be process learning through joint efforts with governments and partners in anticipatory action programming.

Such learning will be divided into two streams:

1. Independent agency M&E which is natural part of any programme and;
2. A multi-stakeholder and multi-disciplinary impact assessment of the CERF pilot to develop a cohesive narrative on the impact of this intervention across the different UN agencies, contributing disciplines and implementing partners.

Learning from this framework should be achieved with the highest possible standards and rigor. While the project committee will discuss the key learning questions in more detail, the following areas of learning have been pre-identified as possible starting points designing M&E as well as an independent evaluation protocol:

- **Timing:** Was the timing of the intervention in relation to peak cyclone season appropriate?
- **Impact:** Did anticipatory actions reduce the impacts of tropical cyclones on beneficiaries? Did they result in improved outcomes?
- **Joint targeting:** Did the common beneficiary database and joint targeting lead to improved efficiencies and more effective assistance provided to beneficiaries by agencies who used this modality?
- **Social inclusion:** Did the most at-risk populations receive timely assistance? Did the most at risk receive the most appropriate assistance to address their needs?
- **Early warning:** How effective is the joint approach and protocol to early warning? Did the content of the early warning messages matter?
- **Localization:** How did the anticipatory action approach engage with the localization agenda? What can be improved?
- **Partnership:** What elements can be learned from the partnership with the Fijian government? How can the current experience and learning support an institutionalization of anticipatory approaches?
- **Actions:** How can the anticipatory action by each agency be further improved?
- **Impact:** What are the socio-economic impacts (including food security, livelihoods, health and protection) to beneficiary populations from the different AA interventions that can be documented? What is the return of investment for the different AA interventions?

Monitoring & Evaluation

Each agency will use its existing monitoring & evaluation systems to collect and track data on implementation progress and outputs achieved. This can be financed through the CERF funds. All findings will be shared with OCHA. M&E questionnaires should be coordinated among agencies to allow for maximum learning, each to evaluate and report on a set of common questions and indicators across all interventions. This may also enable a reduction in the number of interviews required by beneficiaries. Partners can bring in MEAL experts from their respective global and regional level and can draw on recently published guidance to evaluate anticipatory action.¹⁰ Further common questions can be discussed and agreed through the project committee.

Independent Evaluation

Agencies in the framework also agree to participate in an independent evaluation that will aim to build a cohesive narrative on the impact of this CERF pilot. Such support for the evaluation from the agencies may include, but not be limited to:

- Provide feedback and support to develop an evaluation methodology and protocol;

- Collect pre and post disaster data;
- Share beneficiary data (share information on household characteristics without identifiers) or agree on a third party to collect data;
- Support the roll-out of a common surveys;
- Participate in evidence and learning processes;
- Review and provide timely feedback on the analysis and evaluation report.

The project committee will oversee this evaluation and define the scope of the exercise. The evaluation will review the effectiveness of all AA interventions, from the more hardware-focused approaches (distribution of cash, dignity kits, watertight drums), to software-focused approaches (technical assistance and advisory) and process-focused support (early warning systems and communication).

At present the methodology is suggesting three scenarios (due to the late finalization of the Framework ahead of the cyclone season):

- Scenario A: AA is triggered after January 2024 as according to El Nino cyclone season forecast and there is time for a comprehensive baseline data collection;
- Scenario B: AA is triggered before January 2024, with no time for a comprehensive baseline data collection;
- Scenario C: No AA is triggered during the 2023/24 cyclone season until end April 2024.

The overall evaluation will need to be designed to cover three components:

1. Process learning across partners and joint efforts with the Fijian Government;
2. Socio-Economic household impacts (including food security, livelihoods, health and protection);
3. Return for Investment calculations and considerations based on the various AA intervention streams.

The proposed methodology should include quantitative and qualitative data approaches, and possibly use a quasi-experimental difference-in-difference evaluation design. The evaluation protocol will include tools for data collection, schedule and protocol for data collection, sampling frame for data collection, identification of contributing stakeholders, etc.

In the case of Scenario A, a stronger quantitative household data component is expected, given the available time for the process of baseline data collection. However, for Scenario B, a more qualitative process of post-trigger data collection might be needed to accommodate the lack of a strong baseline before the AA was triggered. And for Scenario C, a stronger qualitative process learning focused evaluation might be appropriate.

FAO is providing technical support for this area in connection with the Asia Pacific Technical Working Group on Anticipatory Action, which is co-led with IFRC.

6. Final provisions

Advocacy and public communication

Agencies agree to coordinate, under the leadership of the NDMO and the Resident Coordinator, key messages around the framework, including during the development, activation and learning phase.

In case of an activation of the framework, close coordination facilitated by OCHA will facilitate as a minimum: a joint press release (Annex 11) and mutual amplification of social media stories.

To enhance common messaging and advocacy for anticipatory action, agencies agree to share any visuals (photos, videos) and impact stories with the ad hoc committee as these become available.

Annex 1 – Trigger Evaluation Specification

The Category of the tropical cyclone is included in the forecast from RSMC Nadi, which uses the Australian tropical cyclone intensity scale.

The official forecast published by RSMC Nadi has a maximum lead time of 72 hours. This corresponds to the portion of the overall 120-hour forecast that is used to evaluate the activation trigger. The portion of the forecast with lead times from 72 to 120 hours is not part of the official RSMC forecast and is only being used to evaluate readiness trigger. The full 120-hour forecast is not publicly available, but the truncated 72-hour forecast (*i.e.*, the official forecast) is published¹¹ by RSMC Nadi.

The distance of a cyclone forecast from Fiji is calculated from the “best track” forecast, which indicates the position of the centre of the eye of the storm. A cyclone is deemed to “make landfall” if its track, interpolated linearly between each forecasted point, passes over any point in land in Fiji. It is deemed to pass within 250 km of Fiji if its interpolated track passes over the shape defined as a 250 km buffer around Fiji, calculated using the EPSG:3832 projection. The land in Fiji is defined by the shapefile for Fiji on the Humanitarian Data Exchange¹², which excludes the uninhabited island of Ceva-I-Ra.

Annex 2 – Historical Analysis

Historical Actual Tracks

Based on the historical official best tracks of tropical cyclones since the 1969/70 season, the trigger conditions would have been met by 20 tropical cyclones in the past, corresponding to a return period of **2.7 years**. Of these 20 cyclones, 15 would have met the “*Category 4 or 5 while within 250 km*” criterion, 9 would have met the “*Category 3, 4 or 5 while making landfall*” criterion, and 4 would have met both criteria.

In the past ten seasons (*i.e.*, back to and including the 2013/14 season), the trigger would have been met by the following cyclones:

- Winston 2015/16
- Gita 2017/18
- Pola 2018/19
- Harold 2019/20
- Yasa 2020/21

Note that the above return period analysis is based on the actual tracks of the cyclones, not the historical forecasted tracks, thus is only an approximation of the historical return period of cyclones that would have triggered this framework.

Historical RSMC Nadi Official Forecasts

Based on the historical official 72-hour RSMC Nadi forecasts for Evan 2012/13, Harold 2019/20, and Yasa 2020/21, the action trigger would have been activated on average **70 hours** before the cyclone either first made landfall or passed closest to Fiji.

Historical ECMWF Forecasts

ECMWF hindcasts (available¹³ since the 2006/07 season) were also used to estimate the trigger’s historical performance. Based on these hindcasts, the readiness trigger would have been activated on average **87 hours** before the cyclone either first made landfall or passed closest to Fiji.

Of the 8 cyclones in the dataset who would have triggered the action trigger, the readiness trigger gave advance warning for 7 of them. For the remaining cyclone (Daman 2007/08), the readiness trigger would have gone off at the same time as the action trigger.

The readiness trigger would have given a false alarm (*i.e.*, triggering, but the actual path did not end up meeting the trigger conditions) for one storm (Keni 2017/18).

For the remaining 82 storms in the dataset, the readiness trigger was correctly not activated.

¹¹ Recent official forecast tracks available here: <https://www.met.gov.fj/index.php?page=trackmap>

¹² <https://data.humdata.org/dataset/cod-ab-fji>

¹³ <https://rda.ucar.edu/datasets/ds330.3/>

Annex 3 – Distribution list

Distribution list A

To	
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Annex 4 – DRAFT trigger messages

From: data.science@humdata.org

To: distribution list (see annex)

SUBJECT: Anticipatory action Fiji – Readiness trigger reached

[readiness trigger]

This is an automated message.

Dear Resident Coordinator, dear colleagues,

The readiness trigger of the coordinated anticipatory action pilot for tropical cyclones in Fiji has been reached in line with the 120-hour forecast shared by the Fiji Meteorological Service, operators of Regional Specialized Meteorological Centre (RSMC) Nadi.

Readiness activities in line with the framework document [link] should now commence. CERF funding to UN agencies is being released. OCHA may convene a meeting of the participating partners.

Best wishes and good luck,

OCHA Center for Humanitarian Data

[action trigger]

SUBJECT: Anticipatory Action Fiji – Action trigger reached

This is an automated message.

Dear Resident Coordinator, dear colleagues,

The action trigger of the coordinated anticipatory action pilot for tropical cyclones in Fiji has been reached in line with the 72-hour official forecast shared by the Fiji Meteorological Service, operators of Regional Specialized Meteorological Centre (RSMC) Nadi.

All activities in line with the framework document [link] should now commence. OCHA may convene a meeting of the participating partners.

Best wishes and good luck,

OCHA Center for Humanitarian Data

PS: Please remember that the impact of this pilot will be evaluated and we encourage all partners to document decisions and lessons learned as best as possible during this activation. Finally, please also ensure any visuals, stories, social media is shared with the group.

Annex 5 – Draft Complementary cyclone-path information messages

Below is a sample complementary cyclone forecast information message. The forecast in this sample was fabricated purely for the sample and is not valid.

Subject: Anticipatory action Fiji – Forecast information
Date: Thursday, October 5, 2023 at 12:06:28 PM Pacific Daylight Saving Time
From: OCHA Centre for Humanitarian Data
To: Tristan Downing
Attachments: ATT00001.png, ATT00002.png, distances_2020-12-15T0000.csv

This is an automated message produced by the OCHA Centre for Humanitarian Data.

This email is purely advisory, and does not serve as an official notice for the anticipatory action framework. Official notices of activation are sent by a separate email. This email also does not constitute an official forecast. Under the World Meteorological Organization's Tropical Cyclone Programme, RSMC Nadi is responsible for monitoring tropical cyclones in the region, and their forecasts should be treated as authoritative.

Dear colleagues,

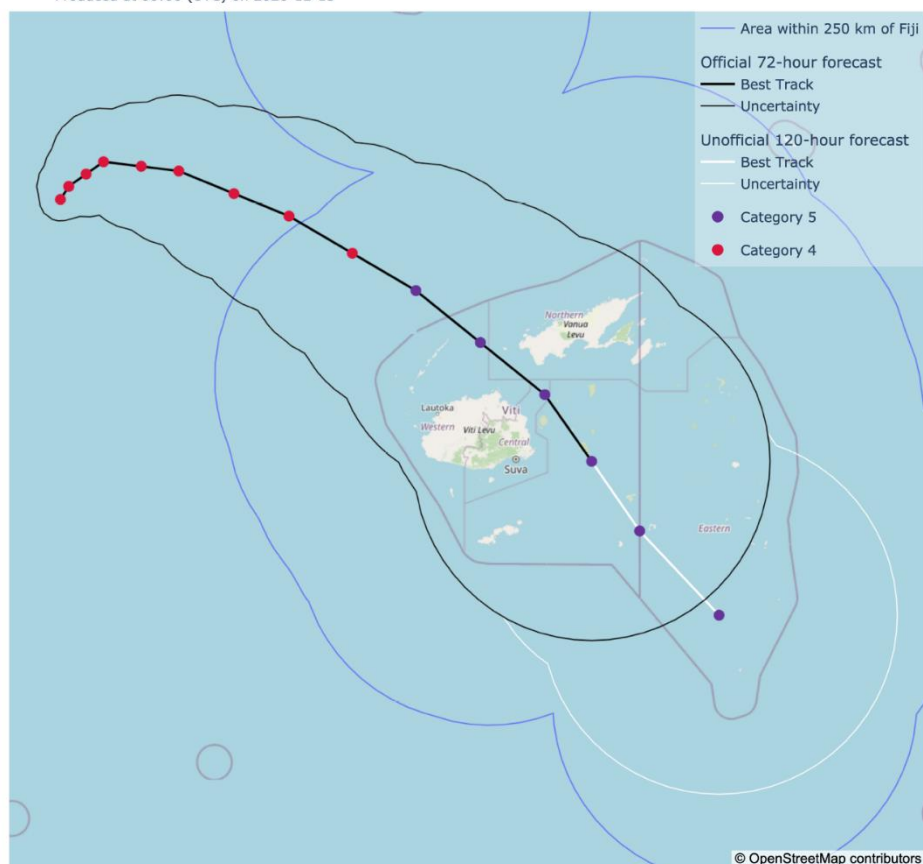
The forecast for cyclone Yasa produced at 00:00 (UTC) on 2020-12-15 has been shared with us by the Fiji Meteorological Service, operators of Regional Specialized Meteorological Centre (RSMC) Nadi.

Readiness trigger: **ACTIVATED**

Action trigger: **ACTIVATED**

A plot of the forecast is shown below. Due to the increased uncertainty of forecasting tropical cyclones beyond 72 hours, this portion of the forecast should be treated with caution.

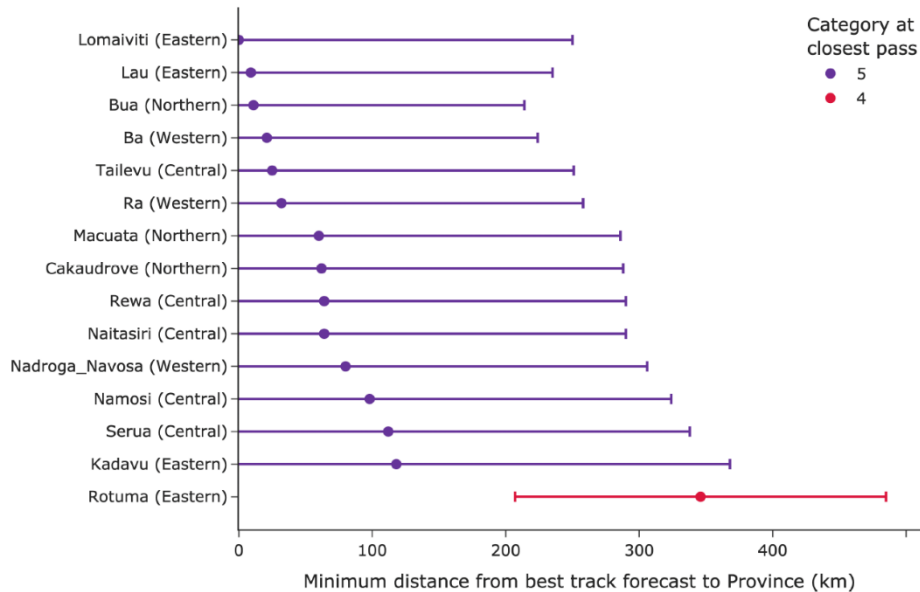
RSMC Nadi forecast for Yasa 2020/2021
Produced at 00:00 (UTC) on 2020-12-15



The plot below shows the distance of each Province to the forecasted cyclone path, including the approximate uncertainty of the cyclone's path at its closest pass to the Province. This data is also included in the attached CSV.

Yasa 00:00 2020-12-15 forecast distances

Error bars estimated based on uncertainty cone of forecast



The code used to produce this alert is available on GitHub [here](#).

Contact the OCHA Centre for Humanitarian Data via Tristan Downing, focal point for Fiji Anticipatory Action at tristan.downing@un.org with any questions or feedback.

Best wishes,

OCHA Center for Humanitarian Data

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Annex 6 – Overview of Activities

Outcome	Intervention	Lead Agency	Government Counterparts	Proposed Activities	Targeted People
The most vulnerable households are able to meet their essential needs	Multi-purpose cash	WFP	Ministry of Women, Children and Social Protection; Ministry of Finance	MPC distribution through DSW	<i>75,000 People</i>
Ensure emergency water, sanitation and hygiene supplies to tropical cyclone affected populations	Non-food items (WASH)	UNICEF	Ministry of Health and Medical Services	- Distribution of emergency WASH kits and tarpaulins - Distribution of MHM pads	<i>20,000 People</i>
Ensure that children return to safe learning environments and are provided with relevant learning resources to continue their education.	Non-food items (Education)		Ministry of Education	- Distribution and installation of temporary learning centres - Distribution and sensitisation of emergency learning materials	
The affected community has access to critical information	Services (Community engagement)		NDMO/ Ministry of Communication	- Mass Media messages distribution - SMS Blasts	
Women and Girls, women with disabilities have access to essential hygiene items	Non-food items (protection)	UNFPA	Ministry of Women, Children & Poverty Alleviation	Women with Disability Dignity Kits – Prepositioned supplies with DPO, to be distributed as preparedness as soon as disaster path is defined	<i>2,350 People</i>
Pregnant women have access to clean delivery kits through health facilities	Non-food items (Health)		Ministry of Health & Medical Services	Distribution of clean delivery kit to visibly pregnant women in the community through community health workers to ensure safe delivery in case they not able to reach health facilities or Health facilities and supplies are destroyed.	
Dignity Kit for women & Menstrual hygiene kits for Young Girls	Non-food items (protection)		Ministry of Women, Children & Poverty Alleviation	Distribution of Dignity kits usually intended for one per household for women aged (15-49) and menstrual hygiene kits one per young girl aged (10-14)	
Decreased damage and loss to crops, tools and fishing gears of	Non-food items	FAO	Ministry of Agriculture and Waterways	Distribution of watertight drums	<i>2,400 People</i>

farmers and fisherfolk					
Decreased damage and loss to boats of fisherfolk	Non-food items		Ministry of Agriculture and Waterways	Distribution of Small boat strengthening kits	
Farmers and fisherfolk are informed that can take locally led actions	Services (Early warning messages)		NDMO/ Ministry of Communication/ Ministry of Agriculture and Waterways	EWM on potential locally led actions	
Decreased damage and loss to shelters	Non-food items	IOM/IFRC	Ministry of Rural & Maritime Development and Disaster Management	Distribution of tie-down kits and tarpaulins, including basic tools and materials for minor works	2,760 People
Evacuation centres management is boosted	Services (Evacuation centre coordination)		Ministry of Rural & Maritime Development and Disaster Management	Coordination, monitoring and support to Evacuation Centre management in support of NDMO	
	Services (Gender and Protection)	UN Women	Ministry of Women, Children and Social Protection	- Gender and protection support and analysis - Deployment of DSW's Community Services Teams	
Cluster coordination	Services (Health)	WHO	Ministry of Health and Medical Services	Cross-cutting support to the health cluster	

Annex 8 – Draft Joint Press Release RC-NDMO

GOVERNMENT AND HUMANITARIAN COMMUNITY IN FIJI LAUNCH ANTICIPATORY ACTION FOR TROPICAL CYCLONE [TC name] - AFFECTED COMMUNITIES TO PREVENT HUMANITARIAN SUFFERING

Suva, [date]

The Government of Fiji and the United Nations activated the Anticipatory Action Framework in Fiji to prevent and mitigate the immediate impact of the forecasted [TC name]. The humanitarian community in Fiji has acted on meteorological forecast data to pre-emptively mitigate and respond to the worst likely impacts on TC-affected communities.

To this end, the Central Emergency Response Fund (CERF) disbursed \$x.x million (xxx FJD) to support the anticipatory actions. With these funds, the United Nations, its national partners and the Fiji government, will support some of the most vulnerable and at-risk communities through humanitarian interventions that will help mitigate the impact of the [TC name].

"The humanitarian system must be as anticipatory as possible, and only as reactive as necessary. AA project aims to complement existing mechanisms rather than undermining community resilience and this should really form the basis of our deliverables" said the Permanent Secretary for Rural and Maritime Development and Disaster Management, Mr. Isoa Talemaibua. According to the Resident Coordinator in Fiji, Mr. Dirk Wagener, "Anticipatory action in the Pacific, especially in Fiji, is not merely a response to crises but a proactive investment in resilience and sustainability. By acting with foresight, we can shape a future where vulnerability gives way to preparedness, and communities thrive amidst challenges."

Recipients of these funds enable UN agencies, through collaboration with Fiji national clusters, led by Government Ministries, to communicate early warning messages, distribute multi-purpose cash assistance and comprehensive relief packages (non-food items, such as: clean delivery kits, drums, xxx).

In addition, essential items, such as hygiene, dignity and health kits are distributed to facilitate access to critical health and protection services, xxx clarify activities. Recognizing the specific impacts of disasters on women and marginalized groups, the allocation seeks to address a broad range of their needs across all phases of the response.

The anticipatory allocation from the CERF in Fiji is the latest example of anticipatory humanitarian action for sudden-onset disasters. Developments in data and predictive analytics make it increasingly possible to anticipate when a disaster is about to strike and take necessary action in advance. This approach offers a potentially more dignified, swifter, and (cost-)effective way to respond to the humanitarian needs of people caught in the global climate crisis.

ENDS